

Patroni Failover vs Switchover — Hands-on Lab

Namespace postgres-ha · cluster acid-postgres-ha · 3 Spilo/Patroni members. Apps must use Service acid-postgres-ha, not a hard-coded pod.

Switchover = planned handoff while the primary is healthy (you pick the candidate). Failover = primary is gone; Patroni elects a replica after the etcd lock expires. Do not start the old primary as a second writer. Never restore backups onto live PVCs during this lab.

Lab 0 — Helper commands used everywhere

```
NS=postgres-ha
roles() { kubectl get pods -n "$NS" -l application=spilo -L spilo-role; }
plist() {
  P=$(kubectl get pods -n "$NS" -l application=spilo,spilo-role=master -o jsonpath='{.items[0].metadata.name}')
  kubectl exec -n "$NS" "$P" -- patronictl list
}
who_primary() {
  for p in acid-postgres-ha-0 acid-postgres-ha-1 acid-postgres-ha-2; do
    v=$(kubectl -n "$NS" exec "$p" -- psql -U postgres -Atqc "select pg_is_in_recovery();" 2>/dev/null) || continue
    if [ "$v" = "f" ]; then echo "$p"; return; fi
  done
}
echo "PRIMARY=$(who_primary)"
roles
plist
```

Lab 1 — Baseline health

```
kubectl get postgresql acid-postgres-ha -n "$NS"
roles
plist
PRIMARY=$(who_primary)
kubectl exec -n "$NS" "$PRIMARY" -- psql -U postgres -c "SELECT pg_is_in_recovery();"
kubectl exec -n "$NS" "$PRIMARY" -- psql -U postgres -c \
"SELECT application_name, state, sync_state FROM pg_stat_replication;"
kubectl get svc -n "$NS" acid-postgres-ha acid-postgres-ha-repl acid-postgres-ha-config 2>/dev/null || kubectl get svc -n "$NS"
```

Pass: one master label, one pg_is_in_recovery=f, two streaming replicas. Record the starting primary name.

Lab 2 — Marker table so you can see the new primary accepted writes

```
PRIMARY=$(who_primary)
kubectl exec -n "$NS" "$PRIMARY" -- psql -U postgres -c "CREATE DATABASE IF NOT EXISTS ha_lab;" 2>/dev/null || \
kubectl exec -n "$NS" "$PRIMARY" -- psql -U postgres -c "CREATE DATABASE ha_lab;"
kubectl exec -n "$NS" "$PRIMARY" -- psql -U postgres -d ha_lab -c "
CREATE TABLE IF NOT EXISTS ha_events (
  id BIGSERIAL PRIMARY KEY,
  event text,
  at timestampz DEFAULT now()
);
INSERT INTO ha_events(event) VALUES ('before-switchover on ' || inet_server_addr());
SELECT * FROM ha_events ORDER BY id;
"
```

Lab 3 — Planned SWITCHOVER

Primary stays reachable. You choose the candidate. Patroni drains WAL, promotes the candidate, old primary becomes replica. patronictl flags vary by version; if --master is rejected try --leader.

```
PRIMARY=$(who_primary)
CANDIDATE=$(kubectl get pods -n "$NS" -l application=spilo,spilo-role=replica -o jsonpath='{.items[0].metadata.name}')
echo "PRIMARY=$PRIMARY CANDIDATE=$CANDIDATE"

# preview (some builds support --dry-run; ignore if unknown)
kubectl exec -n "$NS" "$PRIMARY" -- patronictl list

kubectl exec -n "$NS" "$PRIMARY" -- patronictl switchover acid-postgres-ha \
  --master "$PRIMARY" --candidate "$CANDIDATE" --force

# if that fails, try:
# kubectl exec -n "$NS" "$PRIMARY" -- patronictl switchover --force
```

Lab 3b — Prove switchover

```
sleep 8
roles
plist
NEWP=$(who_primary)
echo "NEW_PRIMARY=$NEWP"
kubect exec -n "$NS" "$NEWP" -- psql -U postgres -d ha_lab -c \
"INSERT INTO ha_events(event) VALUES ('after-switchover writer=' || current_setting('cluster_name', true));
SELECT * FROM ha_events ORDER BY id;"

# old primary must now be in recovery
kubect exec -n "$NS" "$PRIMARY" -- psql -U postgres -Atqc "SELECT pg_is_in_recovery();"

```

Expect: NEWP equals CANDIDATE, old PRIMARY prints t, exactly one master label. Two masters = split-brain — stop and call the instructor.

Lab 4 — Unplanned FAILOVER

Do not run this until Lab 3 is stable (one leader, old primary streaming). Deleting the current primary pod simulates a crash. Patroni promotes a remaining replica. StatefulSet will recreate the deleted pod as a replica.

```
CUR=$(who_primary)
echo "About to delete primary pod: $CUR"
kubect delete pod -n "$NS" "$CUR"

kubect get pods -n "$NS" -l application=spilo -w
# Ctrl+C when three pods are Running

```

Lab 4b — Prove failover

```
roles
plist
NEWP=$(who_primary)
echo "PRIMARY_AFTER_FAILOVER=$NEWP"
kubect exec -n "$NS" "$NEWP" -- psql -U postgres -d ha_lab -c \
"INSERT INTO ha_events(event) VALUES ('after-failover');"
SELECT * FROM ha_events ORDER BY id;"

# recreated old primary should be replica
kubect exec -n "$NS" "$CUR" -- psql -U postgres -Atqc "SELECT pg_is_in_recovery();"
kubect exec -n "$NS" "$NEWP" -- psql -U postgres -c \
"SELECT application_name, state FROM pg_stat_replication;"

```

Expect exactly one f from who_primary. Recreated pod should report t. If the old primary comes back writable, do not write to it — that is split-brain.

Lab 5 — Service still follows the leader

```
kubect get endpoints -n "$NS" acid-postgres-ha -o wide
kubect exec -n "$NS" "$NEWP" -- curl -s http://127.0.0.1:8008/patroni | head
# write through the cluster Service from a replica pod (connects to leader)
kubect exec -n "$NS" acid-postgres-ha-0 -- psql -U postgres -h acid-postgres-ha -d ha_lab -c \
"INSERT INTO ha_events(event) VALUES ('via-service'); SELECT inet_server_addr();"

```

Connecting to host acid-postgres-ha should hit the current primary, not a fixed pod.

Lab 6 — Optional switch back (planned)

```
CUR=$(who_primary)
WANT=acid-postgres-ha-0
echo "switch $CUR -> $WANT (skip if WANT is already primary)"
kubect exec -n "$NS" "$CUR" -- patronictl switchover acid-postgres-ha --master "$CUR" --candidate "$WANT" --force
sleep 8
roles
echo "PRIMARY=$(who_primary)"

```

Lab 7 — Final health

```
roles
plist
echo "PRIMARY=$(who_primary)"
kubect exec -n "$NS" "$(who_primary)" -- psql -U postgres -d ha_lab -c "SELECT * FROM ha_events ORDER BY id;"
kubect exec -n "$NS" "$(who_primary)" -- psql -U postgres -c \
"SELECT application_name, state, sync_state FROM pg_stat_replication;"

```

Do / do not

Do	Do not
Use patronictl switchover for planned work	Promote by editing PGDATA or removing recovery files
Delete one primary pod only for failover drill	Delete all three pods or etcd
Confirm exactly one pg_is_in_recovery=f	Write to a node that still thinks it is primary after failover
Use Service acid-postgres-ha	Hard-code acid-postgres-ha-0 in apps
Wait for old primary to show Role=Replica	Restore pgBackRest onto live PVCs in this lab

Pass: switchover moved the writer to your candidate with zero unexpected second primary; failover after pod delete produced a new single primary; ha_events has before-switchover, after-switchover, after-failover rows; old primaries rejoined as replicas.